# Sound speed and resonance

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use travel time and resonant conditions to interpret sound.

**SUCCESS CRITERIA - I can...**

1. Account for round-trip distance in an echo.
2. Identify resonance as a response near a natural frequency.
3. Use an appropriate wavelength relationship.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

At fixed wave speed, frequency becomes 3 times as large. Compare wavelength and typical diffraction through the same opening.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $d_{echo} = v\Delta t/2$; v = 340 m/s.
Echo time is a round trip. Resonance occurs near a natural mode under suitable damping.

**Worked reasoning:** In an ideal tube open at both ends, the fundamental wavelength is twice its length. For L=0.5 m and sound speed 340 m/s, f=340/1=340 Hz.

**Your turn:** Sketch the fundamental with displacement antinodes at both ends.

**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

An echo returns from a wall. Draw the outgoing and returning path. Why does sound's total travel distance exceed wall range?




**Marzano check - circle:** ① ② ③

## 04 Solve and justify

A sound echo returns after .30 s in air. Find wall range. An open-open pipe fundamental has length .50 m; find$\lambda$=2 L and f=v/$\lambda$.

**Marzano check - circle:** ① ② ③

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Continue down each column.

### 05 Analyze the reasoning

Why does a resonant response not mean that any object can respond strongly to every frequency?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A heat engine absorbs $Q_{hot}$=3500 J and rejects $Q_{cold}$=2450 J each cycle. Find work and efficiency.

**Marzano check - circle:** 1 2 3

### 07 Exit check

A closed-open pipe fundamental has L=.25 m. Useλ=4 L and v=340 to find frequency.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

A 0.85 m tube is closed at one end and open at the other. Usev=340 m/s andλ=4 L for its fundamental. Find the first three allowed ideal resonance frequencies.

**Marzano check - circle:** 1 2 3